



DRDO Netra

More information on India's mass surveillance system, and the ethics needed for implementation. <https://digit.in/jun20-120>



Cybercriminals with CIA tools

Trojans, malware and other hacking tools used by the CIA, that cybercriminals have gotten hold of. <https://digit.in/jun20-121>

FEATURE

implemented on telecom infrastructure in the early 90s. The same system has evolved and is now used to intercept internet traffic as well, by tapping into the infrastructure of internet service providers. Unlike implementations by other countries, the same system can be used for targeted, individual surveillance as well. Many other countries have similar systems in place, and in some countries mass surveillance is simply not covert.

PEGASUS

Pegasus is a software developed by an Israeli cyberweapons community known as the NSO group. The software uses undisclosed vulnerabilities to compromise the smartphones of targeted individuals. The software can be installed on the devices of users through just a text message or a whatsapp forward, without the need for any further interaction. Once compromised, all the information on the device can be tapped, including instant messages, calls and passwords. The data from the microphone or the camera can also be obtained. The NSO group officially claims that the software is available only to authorised government agencies. Cellebrite, an Israeli company has not revealed the full extent of its association with the NSO group, but there have been some employees moving between the two organisations, and both have been known to use similar exploits in their tools. There are some indications that it was Cellebrite that unlocked Syed Rizwan Farook's iPhone for the FBI, a terrorist who was involved in the 2015 terrorist attack in California. The FBI was fighting Apple in court over unlocking that phone. A software with many technical similarities to Pegasus was used to compromise Jeff Bezos' smartphone. In the wake of the novel Coronavirus outbreak, Cellebrite is approaching governments with methods to find out the information needed for contact tracing, without the cooperation of the users, according to a Reuters report.

VAULT 7

The most interesting and recent information on spy tools appeared on WikiLeaks, over a series of disclosures collectively known as the Vault 7 Leaks. All the software in the Vault 7 leaks were being used between 2013 to 2016. At times, the CIA actually repurposed the tools used by cybercriminals for their own use, which provided the additional function of hiding the origin and use of the spyware, if it was to be discovered. These are a collection of some incredibly ingenious and specific software that are used on targeted individuals. For example, Sonic Screwdriver is a software that allows

is a tool that allows wireless routers to be compromised, which means that all the data passing through the router is also compromised. The tool allows loading exploits to execute a variety of man in the middle attacks. Operators can control, monitor and even change the data being transmitted from the router or access point to a machine or device. The attack could be used in public wifi hotspots, or in hotels, or airports. Dumbo is a tool that can be used to compromise webcams or security cams. The related processes to recording videos can be interrupted or stopped. Any recorded footage can also be corrupted. The software is loaded



A screenshot of the CherryBlossom software

an Apple computer to be persistently compromised, by installing malware on the firmware. Even if the firmware is password protected, the malware is stored in the Thunderbolt to Ethernet adapter during boot time. Brutal Kangaroo is a tool that allows snooping on single air gapped machines, or air gapped networks. The attack works by first compromising a machine plugged in to the internet. When a USB device is used on this machine, the software jumps on to the drive. Then, when the USB device is used on a machine on an air gapped network, the malicious software begins its snooping and begins to store data back on the USB drive. When the drive is connected back to a machine linked to the internet, the data is beamed to the agent. CherryBlossom

through a pendrive, and can be used to mask the physical movements of operatives through an area.

The Marble Framework is an obfuscation tool that prevents any forensic investigators from associating malware on a compromised machine to the CIA. A devious approach by the software was to first use text in a foreign language, such as Chinese or Farsi, then obfuscate that language. Investigators would believe that the software was attempting to cover up the use of language, and end up making the wrong conclusion. While all of these might seem really scary, the tools are pretty advanced and not available to the average script kiddie. Only those who may be the individual targets of attacks have to worry about these tools or similar ones. **4**